

# CHARLOTTE LANGE

## Machine Learning for Climate

★ 2000-02-08

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🔗 geronimocharlie

🏠 homepage

## EDUCATION

### M.Sc Environmental Science

#### Universität Freiburg

📅 October 2024 –

📍 Freiburg, DE

In the *Environmental Modelling and Data Science* major, I focus on machine-learning applications to environmental challenges, alongside coursework in statistics, environmental data wrangling and ecological systems.

**1-year ELIZA scholarship (Konrad Zuse School of Excellence).**  
Funding could not continue due to new regulations for non-technical faculties.

### B.Sc. Cognitive Science

#### Universität Osnabrück

📅 October 2018 – April 2023 📍 Osnabrück, DE

In this highly interdisciplinary program, my personal focus lay on Deep Learning, Machine Learning, Neuroinformatics and Neuroscience. Final Grade: 1.2 (GPA 3.8)

**Thesis: "Improving Physical Consistency in ML-driven Radiative Transfer Parameterization for Climate Model Emulation"**  
Supervisors: Dr. Christoph Irrgang (RKI, Berlin), Prof. Dr. Gordon Pipa (IKW, Osnabrück), Grade 1.0 (GPA 4.0)

### A Levels

#### Jacob-Grimm-Schule

📅 August 2015 – July 2018 📍 Kassel, DE

Focus on English and Mathematics. Final Grade: 1.1 (GPA 3.9)  
Receiver of the school's graduate extinction in mathematics and physics. First place with the German team in an international physics competition (AYPT 2016).

## TEACHING

### Lecture: Implementing Artificial Neural Networks with TensorFlow

#### University of Osnabrück

📅 WS 20/21 & WS 21/22 📍 Osnabrück, DE

Co-lectured an 8 ECTS Bachelor/Master course, covering fundamentals of ANNs through state-of-the-art models (e.g., Transformers). Designed lectures and assignments; taught under Prof. Michael Franke (20/21) and Prof. Gordon Pipa (21/22).

### Lecture: Deep Reinforcement Learning

#### University of Osnabrück

📅 WS 20/21 📍 Osnabrück, DE

Tutored an intensive 8 ECTS block course, developing a framework for efficient RL implementation, revising content, producing teaching graphics, and providing coding support. Covered basic RL through trust-region methods; taught by Leon Schmid under Prof. Elia Bruni.

## PRACTICAL EXPERIENCE

#### • Research Assistant

📅 October 25 -

📍 Geosense Lab, Freiburg

Initiated a collaboration between the GeoSense and the AutoML Labs. I aim to develop generalized AutoML workflows for environmental data within a project on global plant functional trait mapping.

#### • Independent Research Assistant

📅 May '23 - July '24

📍 MILA, remote

As part of an interdisciplinary, international project at the intersection of causal machine learning and climate physics, I handled coding, academic writing, method development, and management tasks.

#### • Honorary Tutor

📅 since April '23

📍 SFN, Kassel

At Schülerforschungszentrum Nordhessen (SFN) mentor I students in programming, AI, and climate science. My protégé **Lilly Schwarz** won the 2024 national *Jugend Forscht* interdisciplinary prize with her deep-learning climate-simulation project and is competing in 2025 with a deep-learning model of oceanic currents.

#### • Self-organized Research Internship

📅 Aug - Dec '22

📍 MILA, Montreal

During my stay at the Quebec Institute of AI, I worked on acquiring and preprocessing large-scale climate model data for machine-learning purposes in the context of a joint research project by MILA and Intel. Under Supervision of Prof. David Rolnick.

#### • Self-organized Research Internship

📅 June - July '22

📍 GFZ, Potsdam

During my stay at the Helmholtz Institute for Geophysics in the Geodesy Department (Section 1.3. Earth System Modelling) I explored their research projects and the prospects of applying machine learning. Supervised by Dr. Christoph Irrgang.

#### • DAAD funded Research Internship

📅 Aug - Nov '21

📍 UKZN, Pietermaritzburg

Developing an automated dataset collection system and prototyping a deep learning based individual recognition software. Supervised by Dr. Mandlenkosi Gwetu.

## PUBLICATIONS

Kaltenborn, J., Lange, C., Ramesh, V., Brouillar, P., Gurwicz, Y., Runge, J., Nowack, P., & Rolnick, D. (2023). ClimateSet: A Large-Scale Climate Model Dataset for Machine Learning. *Proceedings of the Thirty-Seventh Annual Conference on Neural Information Processing Systems, (NEURIPS 2023), New Orleans, USA, 10-16 December 2023.*

## SKILLS

Python

TensorFlow

PyTorch

end-to-end machine learning pipelines · distributed computing

Linux (Ubuntu)

setting up and configuring systems · virtual environments · problem-solving

LaTeX

R

## RESEARCH INTERESTS

physics-informed deep learning · theory-guided machine learning · climate model emulation

## LANGUAGES

German

English

French

## HOBBIES

hiking · exploring · poetry · dancing · nutrition-science · reading · creative writing · arts ·

## FURTHER EDUCATION

- International Summer School on Renewable Energies  
📅 August 2019      📍 FH Aachen, Jülich  
  
Completed a certified two-week international summer school on renewable energies.  
Grade: 2.0 (GPA 3.0)

## COMMITMENT

### University

- Student rep. for the appointment committee for the prof. in Neuromorphic Computing  
📅 Jan 2022 - Aug 2022
- Member of the Institute of Cognitive Science's Mentoring Team  
📅 Aug 2020 - May 2022
- Student rep. for the appointment committee for the prof. in Machine Learning  
📅 June 2020 - June 2021
- Elected vice- and president of the student parliament as a member of the Green Party  
📅 May 2019 - October 2020